4.04 Further Vectors				
4.04a	Equation of a straight line	a) Understand and be able to use the equation of a straight line, in 2-D and 3-D, in cartesian and vector form. *Learners should know and be able to use the forms:* $y = mx + c$, $ax + by = c$ *and* $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$ *in 2-D, and* $\frac{x-a_1}{u_1} = \frac{y-a_2}{u_2} = \frac{z-a_3}{u_3} (= \lambda)$ *and* $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$ *in 3-D.* *Includes being able to convert from one form to another.*		F1
4.04b	Equation of a plane		b) Understand and be able to use the equation of a plane in cartesian and vector form. *Learners should know and be able to use the forms:* $ax + by + cz = d$, $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}$, $(\mathbf{r} - \mathbf{a}).\mathbf{n} = 0$ *and* $\mathbf{r}.\mathbf{n} = p$. *Includes being able to convert from one form to another.*	F2
4.04c 4.04d	Scalar product	c) Be able to calculate the scalar product and use it both to calculate the angles between vectors and/or lines, and also as a test for perpendicularity. *Includes the notation* $\mathbf{a.b}$	d) Be able to find the angle between two planes and the angle between a line and a plane.	F3 F4
4.04e 4.04f	Intersections	e) Be able to find, where it exists, the point of intersection between two lines. *Includes determining whether or not lines intersect, are parallel or are skew.*	f) Be able to find the intersection of a line and a plane.	F5

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…	DfE Ref.
4.04g	Vector product	g) Be able to use the vector product to find a vector perpendicular to two given vectors. *Includes the notation* $\mathbf{a} \times \mathbf{b}$. *When the vector product is required, either a calculator or a formula may be used. The formula below will be given:* $\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \times \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & a_1 & b_1 \\ \mathbf{j} & a_2 & b_2 \\ \mathbf{k} & a_3 & b_3 \end{vmatrix} = \begin{pmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{pmatrix}.$ [*The magnitude of the vector product is excluded.*]		Essential content for F2, F3, F5, C6
4.04h 4.04i	Shortest distances		h) Be able to find the distance between two parallel lines and the shortest distance between two skew lines. *For skew lines, the formula* $D = \dfrac{\lvert(\mathbf{b}-\mathbf{a}).\mathbf{n}\rvert}{\lvert\mathbf{n}\rvert}$, *where* $\mathbf{a}$ *and* $\mathbf{b}$ *are position vectors of points on each line and* $\mathbf{n}$ *is a mutual perpendicular to both lines, will be given.* *Either* $\mathbf{n}$ *will be given, or it must be established from given information including by use of the vector product.* i) Be able to find the shortest distance between a point and a line. *The formula* $D = \dfrac{\lvert ax_1 + by_1 - c\rvert}{\sqrt{a^2 + b^2}}$ *where the coordinates of the point are* (x_1, y_1) *and the equation of the line is given by* $ax + by = c$, *will be given.*	F5

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
4.04j			j) Be able to find the shortest distance between a poir and a plane. *The formula* $D = \frac{\lvert \mathbf{b.n} - p \rvert}{\lvert \mathbf{n} \rvert}$ *where* $\mathbf{b}$ *is the position vector c the point and the equation of the plane is given by* $\mathbf{r.n} = p$*, will be given.*